

THE QURAN DECIPHERMENT PROJECT



The Perfect Flow

An Essay on the Sequential Coherence of the Qur'an



A research deliverable of the
Quran Decipherment Project

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The Perfect Flow

A single sustained essay on the Quran from its smallest unit outward — the letter to the phoneme to the word to the verse to the surah to the corpus, then outward again to chronology, world, reception, and future.

Begin at the letter. Not the word, not the verse — the letter, because Arabic writing is a dense, opinionated technology and the Quran is, before it is anything else, a string of graphemes obedient to that technology's rules. The script was acquired from Nabataean cursive around the fourth century of the common era; by the time of the canonical recension in the seventh it had no systematic marking of short vowels, no consistent indication of the glottal stop, and — critically — no mark for the *shadda* that doubles a consonant. Two of the most load-bearing facts about Quranic sound are things the rasm does not write. The *hamza* floats on carriers whose identity is dictated by vowel environment: the *alif that silently carries the hamza* at the head of a word is a consonant in acoustic reality and a placeholder in visual reality, and the tension between the two is already the beginning of the text's interpretive life. The *shadda*, a mark added only in later ḥarakāt systems, governs whether the *Rabb* of Sūrat al-Fātiḥa has one *b* or two — whether the Lord of the worlds arrives with a doubled plosive or a single one — and every counting question about the Quran turns, at some level, on a scribal convention the base script does not encode. This is why the project's methodology spends more pages on anchor values and rule tuples than on any novel claim. *A tool that cannot reproduce 330,709 letters under a disclosed orthography rule is not broken in its novelty — it is broken at the level of the letter.*

The Arabic alphabet of twenty-eight letters contains letters that belong almost to the Quran alone. The letter ẓ (ẓā'), a pharyngealised interdental, is cross-linguistically rare; it occurs with markedly lower frequency than its nearest neighbours ط and ذ, and in the post-classical Arabic world it often survives only in Quranic citation. The classical tradition treated ط with grammatical tenderness, compiling whole monographs on *al-farq bayna al-qād wa-l-ẓā'*, because writers kept confusing the two in manuscript. The Quran's ط does not waver. The ض, the letter Arabs call their own — *luḡhat al-qād* — is the marker that recitation tradition defends most assertively in tajwid manuals. These are not incidental. Letter frequency is a stylometric fingerprint, and the project's cross-baseline letter-frequency analysis — computed on the no-tashkeel orthography with hamza-

carriers folded to ءand ىnormalised to ٱ(rules tuple: *orthography=no-tashkeel; hamza-carriers-folded; basmala=counted-only-in-surah-1; verse-numbering=hafs-kufan*) — places the Quran's letter distribution at $|z| > 2\sigma$ distinct from classical Arabic prose on twelve of the twenty-eight letters. و(waw) is $+53.3\sigma$ above the matched baseline; م (mim) is $+46.8\sigma$. These are not accidents of a small corpus; they are the signature of a declarative-hortatory register against a narrative-historical baseline. The Quran writes the way it writes; the letters themselves tell you so before any word has been read.

And the script tells you what it cannot represent. The shadda that is written in the tashkeel JSON but not in the no-tashkeel JSON is the reason the project carries three parallel corpora and pins every counting tool to a specific one. The flat-text files that a first-draft script would reach for were silently truncated by a MySQL GROUP_CONCAT ceiling at Q 61:5 and Q 40:40 — a failure so clean it could have gone undetected had the anchor values not been locked. The locked anchors are: 114 surahs, 6,236 verses, 77,797 real-word tokens, 330,709 letter graphemes, 19 letters in the basmala, 4 words in the basmala. Every claim in what follows must reduce to these numbers under a published rule; a number without its rules tuple is not yet a claim.

There is a further thing the letter-level picture forces one to acknowledge: the muqattaʿāt. Twenty-nine surahs open with isolated letters — alif-lām-mīm, ḥā-mīm, ṣād, qāf, ṭā-hā, kāf-hā-yā-ʿayn-ṣād, yā-sīn, nūn — set apart from the surrounding prose, unexplained by any classical commentator whose explanation held for the whole tradition. Al-Ṭabarī gathered six competing explanations and declined to choose. Al-Rāzī, more systematic, proposed the divine-name abbreviation theory: that the letters are initials of the Most Beautiful Names; we tested that exact theory and it failed (66 of 99 Names begin with a "luminous letter", but a proper random-14-subset null gives $p = 0.139$; mīm alone drives 136% of the apparent excess, and the excess disappears under control). What does survive — and survives triple nulls — is not al-Rāzī's semantic theory but a statistical one: the letter that opens a muqattaʿāt surah is over-represented *inside* that surah's body, at $\chi^2 = 228.78$ on 78 degrees of freedom, $p < 10^{-15}$. Under a 3-gram letter-Markov surrogate that preserves local bigram and trigram statistics, the Stouffer combined Z across the 29 surahs is $+4.48$, one-tailed $p \approx 3.8 \times 10^{-6}$. Under a positional-gradient null that would have exposed a topical-onset artifact, Q 50's *qāf* quartile pattern $2.37 \rightarrow 4.65 \rightarrow 4.50 \rightarrow 3.56$ peaks in the second quartile, not the first — the signal is surah-wide, not front-loaded, and the topical-artifact hypothesis is dead. Khalifa's framing of this density as a divisibility-by-19 miracle fails (only 1 of 29 surah-opening letter sums is divisible by 19, exactly chance); but the density effect itself is real. The opening letters of a muqattaʿāt

surah are the surah's own alphabet writ large. This is a letter-level finding that needed the letters counted and the null model shuffled before it could be separated from the apologetic overreach that had buried it.

From letter, move to phoneme. Reciting is not writing. The letter ق that the printed mushaf carries is, in the mouth of a reciter, a specific acoustic object — a uvular plosive with pharyngealisation — and the Quran's sonic character is not the arithmetic of its letters but the classes of their articulation. The phonaesthetics agent measured every verse against six overlapping phonetic categories drawn directly from the classical grammarians: *emphatic* (ص ض ط ظ ق ر, Sibawayhi's *mufakhkham* / *musta'liya*), *resonant* (ن م ل), the sonorants and semivowels), *plosive* (ء ب ت د ط ق ك ج, the *shadīd* list), *fricative* (ر ي و, the *shadīd* list), *fricative* (ث ح خ ذ ز س ش ص ض ظ غ ف ه, the *rakhāwa* continuants), *guttural* (ء ه ع ح غ خ), the throat letters, *hurūf al-ḥalq*), and *labial* (ب م ف و, the *shafawī*). The classes overlap deliberately — ط is emphatic AND plosive, ق is emphatic AND plosive — because the phenomena overlap in nature. The global profile of the Quran under these classes is unambiguous: verses are on average 45.5% resonant, 21.7% labial, 21.1% plosive, 16.7% fricative, 16.1% guttural, 8.0% emphatic. These are the numbers the recitation is made of.

Behind these six classes stand the *makhārij al-ḥurūf* — the articulation points — that Sībawayhi catalogued in the *Kitāb* in the late eighth century and that tajwid pedagogy has preserved without material alteration for a thousand years. The throat has three points (hamza and hā' at the back; 'ayn and ḥā' at the middle; ghayn and khā' at the front) — the *hurūf al-ḥalq*. The tongue carries ten points from the uvular qāf through the palatal jīm-shīn-yā', the alveolars, the interdentals. The lips carry four. The nose carries the ghunna, the nasal resonance of *n* and *m* under specific rules. No other sacred text that comes to mind has a phonology this carefully taxonomised — the recitation tradition is a phonetic-laboratory conservation of exactly how each letter of the Quran is supposed to leave the mouth. When our agents split the graphemes into six classes, they were implementing the ancient taxonomy in code. The percentages above are the Quran's acoustic identity, in Sībawayhi's categories, counted.

The question the classical tradition of *i'jāz ṣawtī* — the acoustic inimitability — posed for fourteen centuries is whether sound enacts meaning at the verse level. Does a punishment verse sound harsh; does a mercy verse sound soft? The agent ran seventy-eight topic-by-phoneme-class Welch tests under Bonferroni at $\alpha = 0.05/78 = 6.41 \times 10^{-4}$. The overwhelming answer is that the classical case is local, not global. Punishment verses are NOT emphatic-heavy at a Bonferroni-surviving level; mercy verses are modestly resonant; the sharp, pre-theoretical intuition that the Quran sounds like its meaning holds at

the scale of specific verses and not at the scale of a topic class. What does hold, at a level no statistic could dismiss, is the one perfect onomatopoeic verse. Q 91:14, *fa-damdama 'alayhim rabbuhum bi-dhanbihim fa-sawwāhā* — "So their Lord annihilated them for their sin and levelled them" — contains the hapax root *d-m-d-m*, a reduplicated plosive-labial compression meaning to crush, to pulverise, to rumble-flatten. The verse describes a mountain being reduced to earth; the sound of the verb is that reduction acoustically enacted. Four phonetic markers stack — plosive, labial, reduplicated, hapax — and the only Quranic occurrence sits in the exact verse where the event happens. Around it, Q 69:13 is 47.8% fricative at the blast of the *ṣūr*, the eschatological trumpet; Q 114:5 hides the hapax *yuwaswisu* (the whisperer whispers) inside the surah against whispering; Q 99:1 zalzala-s the earth when it is shaken. Four perfect sound-is-meaning verses, and then the honest verdict: *local truth, global nuance*. Classical *i'jāz ṣawtī* is vindicated as a phenomenon of particular places, not as a systematic law of the corpus. The ear was right; the theorem was not.

Move to the word. There are 77,797 real-word tokens in the no-tashkeel canonical corpus, which under QAC morphology reduce to 4,832 distinct lemmas and 1,642 trilateral roots. These three numbers — 77,797 / 4,832 / 1,642 — are what the Quran is at the lexical level, and the ratios matter: 16 tokens per lemma, 47 tokens per root, 2.9 lemmas per root. Beneath those averages lies a radically skewed distribution. Nearly 42% of distinct word-forms appear exactly once in the Quran — the *hapax legomena*, the lexical one-offs, the words the Quran speaks and then never speaks again. This is a high rate for a text of this length, and the question that matters is not that they exist but where they go. They are placed deliberately. Under a rules tuple of *orthography=no-tashkeel; position=verse-final; segmentation=whitespace*, hapax words are over-concentrated at verse-final rhyme positions at $p = 7.35 \times 10^{-29}$ — a figure so extreme that even aggressive multiple-comparison correction cannot touch it. The Quran's rare words are not scattered; they are rhymed. A word that will never be said again is given the emphatic position in its line, and the classical *saq'* that Arab grammarians heard as the text's acoustic spine turns out, when its positions are counted, to be a rare-word delivery system.

The lexical field of the divine is where this shows most vividly. Fifteen unique divine-name tokens — names that occur nowhere else in the Quran — concentrate in just three verses: Q 59:22, Q 59:23, Q 59:24, the *khawātim* (seals) of Sūrat al-Ḥaṣhr. Eight of those names — *al-Quddūs*, *al-Salām*, *al-Mu'min*, *al-Muḥaymin*, *al-Jabbār*, *al-Mutakabbir*, *al-Bārī*, *al-Muṣawwir* — appear literally nowhere else in the corpus. The three verses contain, in total, 49 words (7²) and 216 letters (6³) — arithmetic coincidences in a canonical Medinan pericope whose content is "the most beautiful names belong to Him", and whose structure matches the content by concentrating those names to a density found

nowhere else in the book. Q 59:23 is rank 1 of 6,236 for divine-name-per-word density (50%). The classical tradition called this the "Greatest Name" passage and the computational infrastructure now confirms, at scale, that its density is in fact unique. Paired against this accumulative mode at the end of the book is a compressive one: Sūrat al-Ikhlāṣ (112) in just four verses houses a single Quranic hapax, *al-Ṣamad*, a name that exists in the Quran exactly once. Two opposite strategies of divine-predication — accumulation at Q 59:22-24, compression at Q 112 — both functional, both unique, one at the seventy-fifth percentile of the mushaf and one at its tail.

Rahma, mercy, appears as a lemma exactly 114 times in the Quran — the count matching the surah total. This coincidence was the project's most exciting early finding. Then the base-rate work was done. A cross-baseline test ran 77k matched-length slices of classical Arabic prose and found that 34.1% of such slices contain at least one lemma with a unique count of 114. The miracle-number for an arbitrary thematic word is a one-in-three event in comparable prose. The Bonferroni-adjusted p for *rahma*-equals-114 in isolation is 1.000. The coincidence is real; the statistical uniqueness is manufactured. The finding was therefore demoted to "base-rate coincidence; held at the lexical-family level" in [findings/phase-b-hypotheses/rahma-114-baseline-rigor.md](#): within the r-ḥ-m family *rahma* is still the only member at count 114, which is a narrow, family-robust observation. The lesson is larger than the word. The McKay denominator — **2,817** — is the number of trilateral root pairs in the Quran with identical occurrence counts at a floor of ten hits per root. Anyone wishing to publish "the Quran balances X against Y at count N" must beat that denominator. Most such publications have not.

A second observation at the word-level: the Quran's self-reference vocabulary. Within the 4,832 distinct lemmas sits a small family of words by which the book names itself — *qur'ān*, *kitāb*, *dhikr*, *furqān*, *tanzīl*, *mathānī*, *bayān*, *ḥikma*, *nūr*, *rūḥ*. Ten words, each carrying a functional identity: the recitation, the written book, the reminder, the criterion between right and wrong, the sent-down, the twin-pairings, the clear disclosure, the wisdom, the light, the spirit. Their combined count within the corpus is in the hundreds, and their distribution is not neutral: *qur'ān* clusters where the text self-instructs; *dhikr* clusters where the text invokes memory; *furqān* where judgement is at stake. Most strikingly, *mathānī* — the twin-pairings — is the self-description the text gives at Q 39:23 (*allāhu nazzala aḥsana l-ḥadīthi kitāban mutashābihan mathāniya*, "Allah has sent down the best of speech, a book similar, paired"), and its root *th-n-y* is the root of the Arabic number two, of the dual grammatical form, and of the verb to double. The Quran is, by its own lexicon, a book of pairings. Every ring structure our algorithm measures is a *mathānī* event. The text gave the name for the phenomenon before any commentator described it and before any computer extracted it.

A third word-level feature the corpus makes visible is the Zipf distribution of the Quran's vocabulary. Log-log fits of rank against frequency across the full lemma list give an exponent $\alpha \approx 1.318$ under the lemma counting rule and $\alpha \approx 0.97$ under the surface-form rule (OLS, $R^2 = 0.975$ at the lemma rule). Heaps $\beta \approx 0.618$ — the vocabulary-growth curve — is unusually high for prose. What this says, in plain terms, is that the Quran is *both* more repetitive at the top (its top twenty or fifty lemmas carry more of the mass than in typical prose) *and* richer at the bottom (its hapax tail is thicker). The text is at once more focused and more varied than the prose it is usually compared with; and its information-theoretic signature is the fingerprint of that double anomaly. At standard counting rule (surface form) the Quran's α is not distinctive against the baseline corpora (0.94-1.07 across Mu'allaqāt, Ibn Ishāq, al-Jāḥiẓ, Bukhari); at lemma rule it is. The counting rule decides the miracle. Which is precisely the point of publishing the rule.

A word-level observation the text makes about itself deserves mention before we move on. In the Quran there are roots for bread and for iron, for camel and for cup, for garden and for gate; there are roots for *seven* and *forty* and *seventy*; there are roots for every limb in the body, for rain and for cloud and for mountain and for sea. What is near-totally absent, to a degree that is striking once looked for, is weapon vocabulary. The word *sayf* (sword) does not appear in the Quran. *Rumh* (spear) does not appear. The combat vocabulary that would saturate a war-epic is not there; where combat is described it is described in verbs — they fight, they kill, they strike — not in the nouns of the weapons. A text in which the root *r-ḥ-m* (mercy) appears hundreds of times and the noun *sayf* appears zero times is making a lexical choice. The silence of the weapon-nouns is one of the quiet shapes of the Quran.

Move to the verse. There are 6,236 of them. The natural unit of the Quran is not the sentence — a grammatical category the classical tradition sometimes resisted — but the *āya*, which means both "verse" and "sign", and whose final word almost always rhymes with its neighbours. The rhyme alphabet is narrow: five letters {ن, ا, م, ر, د} close 90.2% of all 6,236 verses; ن alone closes 50.1%; ل, the second most frequent letter of the Quran, is eleven times under-represented at verse end despite being everywhere in the body. What looks like a simple acoustic choice is in fact a highly constrained combinatorial regime. Verse endings are not where the text relaxes; they are where it decides.

Several verses stage more than rhyme. Q 13:28 — *alladhīna āmanū wa-taṭma'innu qulūbuhum bi-dhikri llāh, alā bi-dhikri llāhi taṭma'innu l-qulūb* — reads, in its root skeleton, {Ṭmn qlb dhkr Alh | dhkr Alh Ṭmn qlb}. Four roots, each exactly twice, arranged in perfect mirror around the caesura, eight of nine stem-bearing tokens participating. Its length-normalised jinas density is 0.889, the highest by this metric in the en-

tire Quran. The content is hearts finding rest through the remembrance of Allah, and the sentence enacts that rest: it circles back upon itself the way a resting heart circles back upon its centre. Classical tafsīr surveyed the verse — Ṭabarī, Rāzī, Zamakhsharī, Qurṭubī, Bayḍāwī, Ibn Kathīr — and none of them noticed the chiasmus. The category under which it belongs, *radd al-ʿajuz ʿalā al-ṣadr* ("returning the end to the beginning"), was founded by Ibn al-Muʿtazz in the *Kitāb al-Badīʿ* around 887 CE; Q 13:28 fits the category perfectly but was not listed as an example in the classical catalogs, which tended to take shorter, more immediately perceptible mirrors. The project's computational extractor finds it by blind search. That is the clean case of classical-vocabulary-meets-computation-al-reach.

Q 18:50 is a different kind of precise. The whole-Quran word-midpoint under one tokenisation falls inside Q 18:50, the sentence *illā iblīsā kāna mina l-jinni fa-fasaqa ʿan amri rabbih* — "except Iblīs, he was of the jinn and disobeyed his Lord's command". That verse is simultaneously the word-midpoint of the entire mushaf AND the only jinn-root mention in all of Sūrat al-Kahf. The anchor word at the book's dead centre is "jinn" in a context that names Iblīs's disobedience. The convergence is not merely numerical; it is thematic. The midpoint of the book is where the created fire-being disobeys the created light-being, and the sentence records both in one line.

Q 91:1-7 stages the same enactment at the letter-count scale. The seven cosmic oaths — by the sun, by the moon, by the day, by the night, by the sky, by the earth, by the soul — have letter counts [12, 14, 15, 15, 15, 14, 12]. A palindrome. Mirrored around the central oath, *wa-l-layli idhā yaqshāhā* ("and by the night when it covers it"). Empirical $p \approx 0.007$ against a length-matched random null. The sun-oath and the soul-oath have the same length. The moon-oath and the earth-oath have the same length. The cosmological frame is acoustically symmetric around the concealing night — the Quran's own oath-list is built as a mirror, and the mirror survives a null model. The sister structures are Q 81:2-8 (the Takwīr apocalypse cascade) and Q 37:127-133, each length-7 palindromes of their own. And at Q 19:20-24, the Annunciation scene in Sūrat Maryam, the word-count across Gabriel's utterance, Mary's bewildered reply, and Gabriel's response forms another ring. The Annunciation palindrome is the one people tend to hear with their ears before they count; the sun-oath palindrome is one they hear only once the count has been run.

Rhyme-break verses — verses whose last word violates the surah's otherwise stable rhyme — turn out, with striking regularity, to be doctrinally loaded. Sūrat Maryam's rhyme breaks at vv 34-40 and vv 88-93 fall precisely on the two Christological rebuttals, the surah's polemic against the "Allāh has taken a son" claim. A rhyme-break is a prosodic siren; the text uses it to mark a moment where the acoustic contract is suspended for the sake of saying something that cannot be said in the ambient music. It is the verse

equivalent of a handwritten note in a typed letter. Five modes emerge from the full catalog: Mode A singleton doctrinal (Maryam); Mode B head-block; Mode C tail-block; Mode D dialogue-forced; Mode E legal-content. The taxonomy did not exist in the computational literature before this project.

Near-identical verse pairs are themselves a verse-level phenomenon that the classical tradition named. Al-Zarkashī's *nawʿ* 52 in *al-Burhān fī ʿulūm al-Qurʾān* defined *al-mutashābih al-lafẓī*: the pairs of verses that occur twice in the Quran, separated by many surahs, with small variations. His most famous cited example is Q 2:58 and Q 7:161 — the pair about entering the city and saying *hiṭṭah* (lay down our sins). The two verses are almost identical in Arabic but differ in a handful of particles and inflections, and al-Zarkashī made a strong thesis of this: *every difference encodes meaning*. Our *mutashābih-lafẓī* agent tested the thesis at scale. Extracting all verse pairs with overlap ≥ 0.80 , we found 265 near-identical pairs, 95 at overlap exactly 1.0 (88 byte-identical). The strong form of his thesis — that *every* difference encodes meaning — fails at the margin, because some differences are pure prosodic choice. The weak form — that the differences are interpretively load-bearing in a large fraction of cases — robustly vindicates: 7 of 10 particle and inflection differences in sampled pairs carry classical-tafsīr interpretations. And the computational extractor, running blind over the whole corpus, independently re-discovers al-Zarkashī's own cited Q 2:58 ↔ Q 7:161 pair. The fourteenth-century scholar selected his example from literary intuition; the twenty-first-century algorithm re-finds it from search. *The tradition's pair-catalog is computationally recoverable, and the tradition's pair-interpretation thesis is computationally testable.*

Two further verse-level observations shape the rhetoric of the book. The Quran asks 830 rhetorical questions and answers fewer than fifty. Once every 7.4 verses the text confronts its reader with a question, and the overwhelming majority of those questions are left for the reader to answer. The top formulas are the bare interrogative prefix *ʾi-* (218 occurrences), *kayfa* + verb (79), *a-fa-lā* + verb (45), and the vocative accusation *a-lam tara / a-wa-lam yaraw* ("do you not see? / have they not seen?", 53). Ring centres host questions three times in five, specifically as accusations. And the Quran is radically dialogical: 1,620 quotation-events, one every 3.8 verses. The top speakers are Qul (the divine imperative, 332 occurrences), Moses (184), and "those who disbelieve" (148). Pharaoh is the only speaker in the Quran who self-deifies. Iblīs's prostration-refusal is a single event told four ways across the surahs. Hapax speakers include an ant, a hoopoe, skin, and hands-and-feet: the Quran licenses bodies and creatures to speak at the resurrection, once, in verses that do not repeat. The speech grammar itself carries a doctrine.

Move to the surah. There are 114 of them. The number itself — $114 = 19 \times 6$ — is the anchor for an entire industry of numerological claim and will be audited before this essay closes. But take the surahs first as literary units, because as literary units they are what the Quran primarily offers to a reader. Al-Fātiḥa, the Opening, is seven verses long and operates at entropy extrema: it is rank 5 of 6,236 for lowest letter-entropy in the entire Quran. Al-Ikhlāṣ, four verses long, is rank 1. An-Nās, six verses long, is rank 4. The last three surahs and the first one operate at the book's minimum informational complexity — the book is framed, at both ends, by passages compressed close to the limit of Arabic. Al-Fātiḥa's v5 iltifāt pivot — *īyyāka na 'budu wa-īyyāka nasta 'īn*, "You alone we worship and You alone we ask for help" — sits at metric exact centre: the surah is 29 words across its seven verses, and v5 stands at word 13 | 4 | 12; it is 143 letters, with v5 at letter 61 | 19 | 63. The nineteen-letter pivot is literally identical to the nineteen letters of the basmala. The hadith Qudsī that says "I have divided the prayer between Me and My servant, half for Me, half for My servant" is structurally literal, and the pivot that halves the surah is exactly a basmala long. Al-Fātiḥa has 6 doubled lemmas out of 23 (26% tiktār density — no other surah this dense in repetition), its 18 roots cover 6.4% of all Quranic content-root mass, and its total abjad value = $10,147 = 73 \times 139$, where 139 is Al-Fātiḥa's own letter count.

A seventh-verse surah is an unusual unit for the opening of a 330,709-letter book, and the ratios make its unusualness apparent: Al-Fātiḥa has 29 words, about 139 to 143 letters depending on the basmala policy, 23 distinct lemmas, 18 roots. Those 18 roots cover 6.4% of all Quranic content-root mass — nearly a fifteenth of the Quran's lexical core stated in seven verses. Only Al-Fātiḥa among 114 surahs reaches a 26% tiktār density (doubled lemmas / total lemmas); six lemmas double themselves in the short space, three of them divine-tier (Allāh, al-Raḥmān, al-Raḥīm) and three of them human-tier (*īyyāka*, *ṣirāṭ*, *'alayhim*). The opening surah is designed as a maximum-density concentrate of the book's theological vocabulary; its low entropy is not poverty, it is concentration. And the abjad arithmetic — total value $10,147 = 73 \times 139$, where 139 is the surah's own letter count under one policy — is the sort of arithmetic result one would consider a coincidence if the number on the right were not the surah's own letter count. The factorisation is not an apologetic claim (the 73×139 product is fixed by the letters chosen) but it is an unusual internal self-consistency: the surah's abjad total equals its letter count times a prime. No other surah exhibits exactly this identity.

Al-Baqarah, the longest surah, is the one in which every major structural claim converges. Verses 131-144, the Abraham-to-qibla pericope, score $z = +9.69$ on the ring-audit — the strongest ring in the Quran by a wide margin, surviving Bonferroni at $\alpha = 0.05$ across a 57,996-window family. Mathias Zahniser described this ring on literary grounds

in 1991; Farrin in 2014 strengthened the description; our extractor rediscovers it without being told where to look. The pairs driving the ring are *slm* (submit), *ḥnf* (incline), *millat*, *dīn*, *rbb*, and the proper names Ibrāhīm, Ismāʿīl, Yaʿqūb. Inside this ring sits Q 2:133, one of only twelve Quranic verses with exactly 114 letters — a small-number coincidence at the surah count embedded inside the surah's central ring. And Q 2:149-150, the twin-opener qibla-directive pair, is one of only two consecutive twin-opener verse sequences in the Quran. The other is Q 59:22-23, the Khawātīm al-Ḥashr. The two densest divine-disclosure sites in the book share a rare structural feature at the per-verse level.

Maryam, Sūrah 19, is three-way loaded at every Christological pivot. Verses 34-40 and 88-93 carry a rhyme-break, an iltifāt cascade, and the Christological polemic; form enacts doctrine at three independent structural layers simultaneously. The rhyme-break × iltifāt co-location is NOT a corpus-wide pattern ($\chi^2 = 0.74$, $p = 0.39$), which is exactly what makes Maryam's triple-convergence a local over-determination: the surah stacks devices where the tradition stacks argument, and the stacking is visibly hers. Maryam is also the unique #1 host of *ar-Raḥmān*; Sūrah 19 contains 16 of the ~57 Raḥmān occurrences in 1.57% of the book's verses, a 17.9× density factor. Fifty-five of fifty-seven Raḥmān occurrences are Meccan (96%). Maryam is the only surah organised as a labelled book-excerpt (5× *udhkur fī l-kitāb*, "mention in the Book", exclusive to Sūrah 19). Raḥmān in Maryam is polemically weaponised: absent from the first Christological polemic (where the claim is "Allāh cannot take a son") and present 4× in the second (where the claim is "Raḥmān has taken a son"). The devotional name of the first Meccan surahs becomes, in Maryam, a defence of monotheism in its dispute with the Christology of the Byzantine Arabian periphery.

Al-Kahf, Sūrah 18, is "the middle of the Quran" by five independent methods. Word-midpoint falls in Q 18:50 (one tokenisation) and Q 18:77 (another); letter-midpoint at Q 18:73; Al-Kahf holds the longest alif monorhyme in the Quran (110 of 110 verses); two Bonferroni-surviving sub-rings — the Dhū'l-Qarnayn journey 83-91 ($z = +5.19$) and the Moses-Khiḍr 60-82 ($z = +2.28$) — both pivot on J=1.000 refrain pairs; the *k-h-f* (cave) root is 6/6 in Sūrah 18, a perfect inclusio; the cross-surah rhyme link with Al-Jinn is the densest pair in the book (27 joint occurrences, three rare fāṣilas exclusive to these two surahs). And Q 18:50 is simultaneously the whole-Quran word-midpoint AND the only jinn-root mention in Al-Kahf. The classical reading of Al-Kahf as four trials — faith, wealth, knowledge, power — finishing with the Dajjāl-protection ḥadīth, turns out to sit on a formal scaffolding in which the Moses-Khiḍr and Dhū'l-Qarnayn three-act templates both end with WALL construction. The middle of the book is built twice, at two narrative layers, as a wall.

Ar-Raḥmān, Sūrah 55, encodes the classical four-part tafsīr division in its refrain rhythm. The refrain *fa-bi-ayyi ālā`i rabbikumā tukadhdhibān* ("So which of your Lord's blessings would you deny?") occurs 31 times. The refrains partition as 8 + 7 + 8 + 8 across creation / hell / paradise-1 / paradise-2 — the canonical four-section division the classical exegetes drew on thematic grounds. All three inter-section boundaries fall on refrains (v30, v45, v61). Hell is one-refrain short — an "eschatological deficit" the text itself performs. And the epithet *dhū l-jalāli wa-l-ikrām* ("possessor of majesty and honour") appears exactly twice in the whole Quran, both in Sūrah 55 (v27 Face-of-Lord, v78 Name-of-Lord); inclusio at the epithet level brackets every refrain, every hell-verse, and every paradise. The phonetic paradox resolves: the surah body is 14.5% plosive (below corpus mean); the refrain is 36.8% plosive (2.4× corpus mean); the surah performs, iconically, a soft enumeration interrupted by a hard accusatory question. The tafsīr partition is recoverable from refrain rhythm alone. The classical scholars had divided the surah this way for a millennium; the division was inscribed in a measurable property of the text they were reading.

Āyat al-Kursī (Q 2:255) deserves a verse in its own sentence because it sits inside Al-Baqarah not as part of the Abraham ring but as its own self-contained architectural event. The Throne Verse has 189 letters ($189 = 27 \times 7 = 3^3 \times 7$), 50 words; its ten grammatical colons (*jumal*) form an inside-structure where J1 and J10 both have exactly 14 letters (outer frame), and J3 and J8 have abjad values that mirror to within 23 of ~2,000. Its rhetorical-question centre at J5 — "who is there that can intercede with Him except by His permission?" — sits at the verse's letter-midpoint. The verse operates in apophatic-kataphatic hybrid mode: it denies (no slumber, no sleep, nothing escapes His knowledge) and affirms (the Throne extends over heavens and earth) in alternation. *Al-Ḥayy al-Qayyūm* ("the Living, the Self-Subsisting") — the divine name-pair that opens the verse — appears in exactly three Quranic verses (Q 2:255, Q 3:2, Q 20:111), forming a cross-Quran triptych of the Greatest-Name tradition at three different compression levels. Against Khawātīm al-Ḥaṣhr's pure kataphatic eight-name-octet, Āyat al-Kursī is the apophatic pole: same structural role (ring centre of a pericope), opposite rhetorical device. The two are a diptych, and the classical "Greatest Name" tradition that elevated both passes computational scrutiny in both halves.

Ash-Shu'arā', the Poets, is built on eight paired prophetic seals — the "truly your Lord is the Mighty, the Merciful" refrain that closes every prophet-pericope in the surah. Al-Ḥadīd, Iron, contains a polarity quartet at v3: *huwa l-awwalu wa-l-ākhiru wa-l-zāhiru wa-l-bāṭin*, "He is the First and the Last, the Outer and the Inner". This is the unique four-name divine stack compressed into a single verse; paired-opposites analysis finds Q 57:3 stacks two Bonferroni-significant opposition pairs (*awwal/ākhir* + *zāhir/bāṭin*), the

compressed *muqābala* type-case named already by al-Zamakhsharī. The much-cited "iron-abjad" claim — that *al-Ḥadīd* = 57 (its surah number) and *ḥadīd* = 26 (iron's atomic number) — is arithmetically correct and table-invariant between mashriqi and maghribi abjad (the six letters ا ب ج د ه ز sit inside the 1-400 range where the two tables agree), but the deeper test classifies it as survivor bias: with 114 surahs and many divisibility rules in reach, coincidence of surah-name-abjad with surah-index is a 1-in-6 or better base-rate event. The iron-miracle specifically is rejected; the divine-attribute quartet at Q 57:3, and Q 57:1-6 as the second of the seven Musabbiḥāt, stand.

Al-Ḥashr, Sūrah 59, holds the densest divine-name cascade in the book at its seals (59:22-24), already visited above; it is also the second of the two surahs (with Al-Baqarah) to contain a twin-opener consecutive pair. Al-Ikhlāṣ — four verses, single hapax *al-Ṣamad*, total abjad value near the round 1000 in the mashriqi table — is the book's shortest statement of doctrine and its densest informational compression. Al-Falaq (113) and An-Nās (114) complete the three-surah entropic frame. Al-Fātiḥa and An-Nās share exactly three roots — Allāh, Rabb, Malik — which are the three classical theistic titles: God as named, God as nurturer, God as sovereign; a clean theological bracket around the whole Quran. Al-Falaq names one Lord-title against four evils; An-Nās names three Lord-titles against one evil — an inverse scaling in consecutive surahs. And all three are bound by *qul* as their sole shared word, a triple imperative that forms the book's last instruction.

Al-Ḥashr, mentioned already for its seals, is also the surah in which the twin-opener cascade (Q 59:22-23) mirrors the only other consecutive twin-opener pair in the Quran — Q 2:149-150 inside the Abraham-qibla ring of Al-Baqarah. A "twin-opener" is a pair of consecutive verses that begin with the same phrase (here, *wa-min ḥaythu kharajta* in Al-Baqarah and *huwa llāhu lladhī* in Al-Ḥashr), a rare prosodic device in the Quran. There are only two such consecutive pairs in the entire book, and they sit at the two densest divine-disclosure passages: the qibla-change pericope and the Greatest-Name seals. A small-number coincidence, counted, becomes a structural alignment: the two moments the book makes its identity most intensely explicit share a formal device the rest of the corpus does not use.

The last three surahs — Al-Ikhlāṣ (112), Al-Falaq (113), An-Nās (114) — operate as the book's entropy-frame. Al-Ikhlāṣ is rank 1 in the entire Quran for lowest letter-entropy: four short verses of dense monotheistic doctrine, saying what God is not (begotten, begetting, having-any-equal) and saying what He is (*al-Ṣamad*, the Quranic hapax that appears literally once, here). Al-Falaq names one Lord-title (the Lord of the daybreak) against four evils (the evil of what He created, the evil of darkness-when-it-settles, the

evil of blowers-on-knots, the evil of an-envier-when-he-envies). An-Nās reverses the ratio: three Lord-titles (Lord of mankind, King of mankind, God of mankind) against one evil (the whisperer, *al-waswās*). One-to-four becomes three-to-one; the two *mu‘awwid-hatān* surahs are inverse scalings of the same rhetorical template. All three last surahs share the single word *qul* ("say") — the imperative that sends each of them; *qul* is the only lemma shared across all three. And Al-Fātiḥa and An-Nās — first and last surahs — share exactly three roots: Allāh, Rabb, Malik. The three classical theistic titles (God as named, God as sustainer, God as sovereign) frame the whole book, precisely because those three are the three lexical items the opening and closing surahs have in common. The book's theological bookends are cleanly computable.

Move to the corpus. Take the Quran whole. It names itself, and its self-naming is exactly the methodological challenge. There are at least ten words the Quran uses as its own name: *al-Qur‘ān* (the recitation), *al-Kitāb* (the book), *al-Dhikr* (the reminder), *al-Furqān* (the criterion), *al-Tanzīl* (the sending-down), *al-Mathānī* (the twin-pairings), *al-Bayān* (the clear-disclosure), *al-Ḥikma* (the wisdom), *al-Nūr* (the light), *al-Rūḥ* (the spirit). Each names a different functional identity. *Al-Mathānī* — the twin-pairings — deserves special mention because the word names, in advance, exactly the mode our agents measured. The root *th-n-y* means to double, to pair, to repeat; it is the root inside *math-nā* (two-by-two) and *ithnayn* (two). When Q 39:23 self-describes the Quran as *kitāban mutashābihan mathāniya* — "a book, similar, paired" — it is telling the reader that repetition with variation is the book's mode. Our ring-audit, our *mutashābih al-lafẓī* audit, our oath-pair cartography, our palindrome hunter — every Phase-B and Phase-C structural tool is measuring a species of *mathānī*. Remarkably, Q 39:23 and Q 13:28 share two roots — *qlb* (hearts) and *dhkr* (remembrance) — so the verse that names the Quran's *mathānī* property uses the exact vocabulary of the verse that exemplifies *mathānī* most perfectly. *The text names its own structural principle using the lexicon of its purest example.*

The Quran also describes its own architecture — the thirteen-layer system our agents decomposed — using vocabulary the tradition has always known. *Muḥkam* and *mutashābih*: clear and ambiguous verses (Q 3:7, which declares its own structural axis). *Nāsikh* and *mansūkh*: abrogating and abrogated (a classical hermeneutical category, carefully treated by al-Zuhrī and Ibn al-‘Arabī, defining a dynamic the text applies to itself). *Asbāb al-nuzūl*: the occasions of revelation, which locate verses in the life-events of the Prophet. *Makkī* and *madanī*: the twofold chronological classification. *Itfāt*: the pronoun-shift that al-Suyūṭī catalogued in *al-Itqān* naw‘ 58, which our agent finds at 70.8% baseline enrichment across the corpus, rising to 89.6% for prophet-pericopes (z = +9.37). *Tarṣī‘*: studded-inlay of parallel phrases. *Saj‘*: the rhymed prose whose fāṣila system our

saj-rhyme agent mapped across 6,236 verses. *Jinās*: root-level wordplay, which our agent scores at 0.889 for Q 13:28 and 0.823 for the Joseph peak window. *Tikrār*: repetition, whose distribution the refrain-of-Ar-Raḥmān analysis measured. *Munāsabat*: the thematic coherence between consecutive surahs, which al-Biqā'ī wrote 22 volumes to describe. Each of these Arabic terms names a structural phenomenon; each has a classical operational definition; each was written about for centuries before it was ever computed. The project's structural apparatus is not a Western import; it is the classical vocabulary made measurable, and it is measurable because the classical vocabulary was precise to begin with.

Beyond self-naming, the Quran makes challenge-verses — *mu'āriḍa* verses — that dare the reader to produce "a sūrah like it" (Q 2:23, 10:38), "ten surahs like it" (Q 11:13), or a *ḥadīth* like it (Q 52:34). These are the classical grounds of *i'jāz*, the doctrine of inimitability. We do not test the challenge on its own terms; the challenge is a theological claim whose rules are not pre-specifiable. What the computational layer can say is narrower and more precise. The thirteen-layer architecture — surahs; verses; words; roots; phonetic classes; rhyme; rings; rhetorical questions; quotations; muqatta'āt; oath-clusters; divine names; hapaxes — all of it holds together. Each layer constrains the next. The honest baseline that any apologetic must beat is the McKay denominator of 2,817 equal-count root pairs at a floor of ten. A "miracle" that publishes ten balanced pairs without disclosing the denominator is publishing one sample from a distribution it has hidden. That is the structural failure every popular numerological tradition has shared, and the structural accountability this project was built to install.

There is one more verse-level phenomenon to record before chronology is taken up, and it concerns the Quran's paired-opposites network — the *muqābala* or antithesis structures the tradition named but never systematised. Under paired-opposites analysis at topic level, three pairs score Bonferroni-surviving enrichments: heaven/earth (same-verse co-occurrence at $p = 1.8 \times 10^{-190}$), life/death ($17.6 \times$ enrichment), *dunyā/ākhirā* (this-world / next-world at strong enrichment). A novel pair — hidden/manifest, *bāṭin / ṣāḥir* — comes in at $26 \times$ enrichment ($p = 6.9 \times 10^{-8}$), not previously catalogued as a systematic Quranic *muqābala*. Q 57:3 stacks four Bonferroni-significant opposition pairs in one verse (*awwal / ākhir, ṣāḥir / bāṭin*), the densest antithesis verse in the book. Conversely, mercy / wrath — which popular piety often cites as a quintessential Quranic antithesis — *fails* same-verse testing. The Quran stages mercy and wrath across verse-alternation, not within-verse fusion. Al-Zamakhsharī's *Kashshāf* named *muqābala* as a rhetorical device a thousand years ago; the computational audit confirms the device is

real, identifies which specific pairs drive the signal, and falsifies the pair that popular reading most often attributed to the category. The tradition's category survives; the popular example of it does not.

Move to chronology. The Quran has an order on the page — the canonical mushaf order, which is not chronological — and the classical tradition preserves a chronological order through *asbāb al-nuzūl*. Theodor Nöldeke in 1860 divided the Meccan surahs into three stylistic phases and added Medinan as the fourth. Under the four-phase schema, verse length climbs monotonically: Early Meccan 18.5 letters per verse, Middle Meccan 38.7, Late Meccan 66.0, Medinan 79.9 (ANOVA $F = 210$; Cohen's $d = +1.87$ between Meccan and Medinan). The climb approximately doubles each phase. Sadeghi's 2011 stylometric paper found the same ramp; Al-Suyūṭī's *al-Itqān* naw' 9 already diagnosed it five hundred years earlier. The Quran gets longer per-verse as it ages. And the proper name *Muḥammad* appears exactly four times in the Quran — Q 3:144, 33:40, 47:2, 48:29 — and all four are Medinan. Eighty-six pre-Hijra surahs never name the Prophet by his proper name; they address him as "the Messenger", "the warner", "you". The Hijra inscribes itself grammatically. *Rabb* — Lord — is the only frequent root that declines chronologically (negative Spearman ρ , densest in Early Meccan, thinning by Medinan); every other frequent root is flat or increases. The theological direction of travel is from Rabb-address to community-law. The vocative *yā ayyuhā alladhīna āmanū* ("O you who believe") occurs 89 times in the Quran and all 89 are Medinan; Meccan surahs do not address a constituted community. The Meccan Quran asks metaphysics; the Medinan Quran asks law. K-means clustering on surah root vectors recovers the Meccan/Medinan label at 97%/89% — the traditional chronology is algorithmically recoverable from root vocabulary alone, which is independent evidence that the tradition's period-labels capture real stylistic differences even where their specific dates for individual surahs may be debatable.

One more chronological observation belongs here. The vocative *yā ayyuhā alladhīna āmanū* — "O you who believe" — appears exactly 89 times in the Quran, and every single occurrence is Medinan (89 of 89; zero Meccan occurrences). This is as clean an asymmetry as the corpus produces: Meccan surahs do not address a constituted community because, chronologically, there was no community to address. The Hijra inaugurates the vocative. Similarly, the seven *Musabbiḥāt* surahs — surahs that open with a form of the root *s-b-ḥ* (to glorify) — are Surahs 17, 57, 59, 61, 62, 64, 87. Our surah-boundaries agent extracted exactly these seven by rule (first-word root = *s-b-ḥ*); the classical catalog also names exactly these seven. The computational and the classical extraction agree perfectly. And the five *Qul* surahs — not four, as popular tradition often says, because Sūrah 72

(Al-Jinn) also opens with *qul*. Popular piety silently filters Al-Jinn out of the *Qul* set; the lexical rule does not. That is one of the small catechism corrections the counting produces.

Move outward, to world. The Quran sits in a specific world-historical context, and it holds positions that prefigure, in various ways and at various distances, modern moral consensus. Q 49:13 — *yā ayyuhā l-nāsu innā khalaqnākum min dhakarīn wa-unthā wa-jaʿalnākum shuʿubān wa-qabāʾila li-taʿārafū* — "O mankind, we have created you from a male and a female and made you peoples and tribes that you may know one another" — is a statement of racial universalism made in the seventh century at a time when world religions remained ethnically bounded. Q 2:256 — *lā ikrāha fī l-dīn* — "there is no compulsion in religion" — is a statement of religious-liberty doctrine at the beginning of a polity. Q 5:32 — "whoever kills a soul ... it is as though he has killed all mankind; whoever saves it, as though he has saved all mankind" — is the verse most cited in the post-9/11 public sphere as the Quran's answer to terrorism, and the citation has become a minor genre of its own. Q 30:22 names human language-and-colour difference as signs of the divine; it is read in the late twentieth century after the findings of human genetics, which established that the genetic differentiation between "races" is smaller than the differentiation within. Q 51:47 — *wa-l-samāʾa banaynāhā bi-ʾaydīn wa-innā la-mūsiʾūn* — "we built the heaven with might, and we are surely its expander" — is read, post-Hubble, as a pre-echo of cosmological expansion; the word *mūsiʾūn* carries both "wide-givers" and "those who expand". Q 21:30 says the heavens and earth were once one mass and we separated them; it is read, post-Big-Bang, as singularity-and-separation. These readings are legitimate and they are also honestly critiqued. The Bucaille tradition — Maurice Bucaille, 1976 onward — systematised "scientific *iʿjāz*" and was accused, correctly, of retrofitting: the verses were read through the lens of theories the author already held, and the theories changed between editions. A proper historical reading asks what a seventh-century Arabian listener would have heard and what the text's own grammar will license, not what any later science will mirror. The responsible position is that the Quran's cosmological verses make *claims consistent with* expansion, separation, embryological staging, orbital motion of sun and moon — and those claims are not contradicted by modern findings — but the verses are not *predicting* the findings, because the verses' literary force is evocative and sapiential, not technical. "The Quran is consistent with" is a weaker, more honest claim than "the Quran predicted"; the weaker claim is the one the evidence supports.

A related note on genetic and anthropological reading. Q 30:22 — *wa-min āyātihi khalqu s-samāwāti wa-l-arḍi wa-ikhtilāfu alsinatikum wa-alwānikum* — names language-difference and colour-difference as divine signs, in the same breath and at the same gram-

matical rank. A seventh-century statement that the visible differences between human populations are *āyāt* (signs, verses) of the Creator is unusual in the religious literature of its period; it sits more comfortably beside modern population-genetic findings (which document that inter-population variation is smaller than intra-population variation) than it does beside the racial theories of the intervening centuries. Q 49:13 completes the thought by making the variation functional: "We made you peoples and tribes *that you may know one another (li-ta 'ārafū)*". The purpose clause is social, not hierarchical; the differences are means to mutual recognition, not to ranking. These verses do not require modern science to state anything — the ethical claim is sufficient on its own — but they do age well in its presence. That is the honest form of the Bucaille-tradition claim: not that the Quran predicted modern science, but that certain of its verses, written for their own sapiential purposes in the seventh century, hold their ground when later scientific findings arrive. "Consistent with" is the proper verb; "predicted" is the overreach Bucaille and his imitators did not resist.

Q 51:47, *wa-l-samā'a banaynāhā bi-'aydīn wa-innā la-mūsi'ūn*, closes its verse on the participle *mūsi'ūn* — those who expand, those who make wide. Classical tafsīr (Ṭabarī, Qurṭubī, Rāzī) glossed the word variably as "we have power" (from *wus'* as capacity), "we are providers" (from *sa'a* as abundance), and "we are wideners". The participle carries all three. Post-Hubble, the reading that foregrounds "wideners" — the cosmological expansion sense — gains resonance against an audience that had no such frame. Whether that resonance is a prediction or an emergent consistency is a matter of how one weighs the author's knowable horizon against the word's range. The intellectually honest position is the same as for Q 30:22: the verse means what its Arabic licenses; modern physics does not make it mean more than that; it does not make it mean less either. Q 21:30, *awalam yara lladhīna kafarū anna s-samāwāti wa-l-arḍa kānatā ratqan fa-fataqnāhumā* — "do those who disbelieve not see that the heavens and the earth were sewn together and we unstitched them?" — is the singularity-and-separation echo. The classical commentators read it as a creation-narrative of separating sky from ground. A modern cosmologist reads it as unstitching from a unified state. Both readings are licensed; neither is displaced.

Move to classical reception. This project sits on the shoulders of fourteen named ancestors, and the names deserve to stand. Burhān al-Dīn al-Biqā'ī (d. 1480), *Naẓm al-Durar*, is the first systematic coherence-and-ring commentary in the tradition; Farrin and Cuypers both credit him. Al-Zarkashī (d. 1392), *al-Burhān fī 'ulūm al-Qur'ān*, naw' 52, defined *al-mutashābih al-lafẓī* — the phenomenon of near-identical verse pairs separated across the mushaf with small variations — and our *mutashābih-lafẓī* agent operationalised his fourteenth-century claim, finding 265 near-identical pairs at overlap $\geq$

0.80 and 95 at exact 1.0 (88 byte-identical). Al-Suyūṭī (d. 1505), *al-Itqān fī ‘ulūm al-Qur’ān*, stands at the tradition's encyclopaedic summit: naw‘ 9 diagnosed the verse-length-by-period climb that Sadeghi would quantify five hundred years later; naw‘ 58 catalogued *iltifāt*, the sudden pronoun-shift, which the project's iltifat-catalog agent topic-enriches at scale. Al-Rāghib al-Iṣfahānī (d. ~1108), *al-Mufradāt*, catalogued Quranic rare words — his entries are the direct classical ancestor of our hapax-legomena catalogue. Al-Kirmānī (d. 1106), *Asrār al-Tikrār*, paired 1,100+ cross-surah mutashābih instances. Ibn al-Mu‘tazz (d. 908), *Kitāb al-Badī‘*, founded *radd al-‘ajuz ‘alā al-ṣadr*, the "returning the end to the beginning" chiasmus that our Q 13:28 palindrome inhabits. Al-Ṭabarī (d. 923), al-Zamakhsharī (d. 1144), al-Rāzī (d. 1210), al-Qurṭubī (d. 1273), al-Bayḍāwī (d. 1286), Ibn Kathīr (d. 1373) supply the exegetical backbone; their *tafsīr* is where every verse-level finding lands for sanity-check. Ibn ‘Arabī (d. 1240), *Futūḥāt al-Makkiyya*, founded *‘ilm al-ḥarf* — the science of letters — whose computational test (Bonferroni-surviving for fire→punishment at $p=5.9\times10^{-16}$, earth→creation at $p=8.8\times10^{-12}$, failed for air→wind and water→paradise) found that Ibn ‘Arabī is vindicated where he functions as phenomenologist and falsified where he functions as metaphysician. Al-Būnī (d. 1225), the most numerological of the classical authorities, leaves no footprint.

Khalīl ibn Aḥmad al-Farāhīdī (d. ~791), whose *Kitāb al-‘Ayn* was the first Arabic dictionary and whose prosodic system (the *‘ilm al-‘arūḍ*) codified the sixteen metres of classical Arabic poetry, stands at the very headwater of the Arabic philological tradition; his student Sibawayhi (d. ~796) wrote *al-Kitāb*, the grammar that every subsequent Arab grammarian commented on, taught from, or argued with. When our phonetic classes were defined as emphatic / resonant / plosive / fricative / guttural / labial, the categories were Sibawayhi's. When our abjad arithmetic was computed, the number-letter correspondences were the ones the tradition attributes to al-Khalīl. Al-Farrā’ (d. 822) added the *Ma‘ānī al-Qur’ān* — the grammatical-exegetical atlas that specifies where each grammatical phenomenon in the Quran occurs. These are ancestors whose tools we inherit before we describe our own. The tradition's commitment to precise counting and precise classification began long before anyone in the tradition had an occasion to call it computational.

The fourteen-hundred-year tradition had the full conceptual vocabulary: *munāsabat* (surah-and-verse coherence); *ring composition* under the Arabic name or absent it; *tarṣī‘* (studded inlay of parallel phrases); *jīnās* (root-level wordplay); *tikrār* (repetition); *radd al-‘ajuz ‘alā al-ṣadr* (chiasmus); *saḥḥ* (rhymed prose). What it lacked was the computational infrastructure — the full corpus on a searchable keyword, the QAC morphology on 128,276 Buckwalter segments, the 1,642-root index, the 4,832-lemma list, the null-

model shuffler, the 13.4-million-token baseline corpus of classical Arabic. This project is what happens when fourteen centuries of classical categories meet twentieth-century concordance infrastructure. The formal citation chain spans seven centuries: al-Zarkashī (14th) → al-Biqā'ī (15th) → al-Suyūṭī (16th) → Khalifa (20th) → Philips (1987) → Witztum-Rips-Rosenberg (1994) → McKay (1999) → Cuypers (2007/2015) → Farrin (2014) → Sinai (2017) → this project (2026). Of the forty-two-plus major findings in the current inventory, seventeen are classically catalogued under a known name (the novelty is application to specific verses); fourteen are implicit in the classical tradition but were never quantified; eleven are genuinely novel with no classical category (numerology, Zipf and information theory, chronological metrics, Bonferroni-corrected chiasmus, graph analysis). *The tradition had the words; the project has the numbers; the combination — when it works — is not a displacement of the classical by the computational but their successful marriage.*

Honest corrections must hold the same prominence as successes, because the project's moral spine is that failures are reported as prominently as confirmations. *Raḥma* = 114 was demoted from candidate-headline to family-robust after cross-baseline testing demonstrated that 34.1% of matched-length Arabic prose slices contain a lemma with count 114; the narrower claim — within the r-ḥ-m root family, *raḥma* is the only member at 114 — holds, but the grand claim does not. *Yūsuf* sjn=12 triple was killed by a length-matched Sīra baseline in which 4.5% of matched windows exhibit the pattern; the finding is fully explained by Sūrah 12 being thematically about prison. The 147-triple (*ghayr* / *ilāh* / *jannah* all at 147) was killed by the same baseline — matched Arabic prose has 10,860-13,177 tied root pairs versus the Quran's 16,997; same order of magnitude, pure pigeonhole, the content pattern *lā ilāha ghayruhū* survives as a rhetorical observation. Most of Khalifa's Code-19 apparatus is falsified: only one of 29 surah-opening letter-sums is divisible by 19 (exactly chance); the ALM letter counts do not reproduce under any consistent orthography; the *Allāh* = 2,698 and grand-total = 346,199 pass only if Q 9:128-129 are deleted from the text, which no scribal tradition does; the universal divisibility rule is empirically false. The one non-trivial survivor inside the Code-19 family is the Qaf-50/42/57 triangle, which is a specific and interesting local signal, not a global miracle. Cuypers's Al-Mā'ida whole-surah ring is disconfirmed at the lexical level ($z = -2.06$; more disordered than 99% of random shuffles of its own verses); Farrin's whole-mushaf macro-ring (surah $k \leftrightarrow 115-k$) is disconfirmed at $z = -4.87$, confounded by length sorting. The iron-miracle — the al-Ḥadīd=57 and ḥadīd=26 atomic-number reading — is classified as survivor bias against the baseline of coincidences available across 114 surahs under multiple divisibility rules. No length- ≥ 4 arithmetic gematric progression and no length- ≥ 3 geometric progression exists at the surah-total abjad level. The middle-ayah-

of-Al-Baqarah claim is partially verified: holds at verse-index midpoint, fails at word-midpoint and letter-midpoint. The folk-wisdom that Meccan saj' is denser than Medinan is flat-out false under label permutation ($p > 0.3$ for every rhyme-density metric; the folk intuition operates through verse brevity, not rhyme tightness). Medinan surahs are in fact $1.94\times$ more jinas-dense than Meccan — the conventional rhetorical picture is reversed. *The methodological integrity required to publish these negatives is the same integrity that licenses the positives. A project that can say "raḥma=114 is a base-rate coincidence" is a project whose other numbers you can trust.*

A note on Bayḍāwī, Qurṭubī, Ibn Kathīr. These three names carry the late classical tafsīr apparatus — al-Bayḍāwī (d. 1286), *Anwār al-Tanzīl*, the great compressor of Zamakhsharī minus the Mu'tazilī theology; al-Qurṭubī (d. 1273), *al-Jāmi' li-aḥkām al-Qur'ān*, the legal-exegetical compendium; Ibn Kathīr (d. 1373), *Tafsīr al-Qur'ān al-ʿaẓīm*, whose ḥadīth-first hermeneutic became the reference standard of later Sunni commentary. Our verse-annotation workflow cross-references every substantive finding against these works. Where they have already noticed what the agent is measuring, the attribution is classical; where the agent surfaces a pattern the tafsīr does not name, the novelty is explicit. At Q 13:28, they all comment on the verse's theological meaning; none names its chiasmus. At Q 2:255, all of them name *al-Ḥayy al-Qayyūm* as the Greatest-Name candidate; none quantifies the triptych with Q 3:2 and Q 20:111. At Q 91:1-7, Ibn Kathīr calls the oath sequence magnificent; none of them counts the letters. These are not failings of the classical tradition; they are the openings the computational layer was always going to fill in. The principle is the one the project has repeated: the classical scholars had the hearing; the computational agents have the counting; and when either is missing, the resulting scholarship is incomplete.

Move to future openings. Twenty-two hypotheses are pre-registered in the deep-hypotheses queue and await agent-time: cross-Arabic-baseline tests for the muqatta'āt density effect, for the Zipf exponent, for the Heaps β ; pre-registered divisibility scans under sharper rules; per-century stability tests of every letter and root count across the full manuscript record (Birmingham, Ṣan'ā', Topkapı); a full cross-baseline re-run of the ring-audit against length- and topic-matched pseudo-surahs from the classical Arabic corpus; a formal test of whether *iltifāt* topic-enrichment (70.8% baseline, prophets 89.6%, revelation 83.5%, law 83%, mercy 78%) survives permutation under a stricter speech-act null; a test of whether the Khawātīm al-Ḥaṣhr 8-name hapax cluster finds analogs in any comparable divine-name dense Arabic prose; a falsification test on whether the Moses-Khiḍr and Dhū'l-Qarnayn shared three-act template is a Quranic singularity or a generic near-eastern narrative convention; a *ilm al-ḥarf* audit of every letter under every topic, extending Ibn 'Arabī's phenomenology to all twenty-eight letters; a pre-registered test of

whether the surah-title roots are more confined to their host surahs than random roots of similar frequency. The Quran's eschatological signs are a different kind of future opening: they are the book's claim to resolve, not to test. Preservation-of-text is a matter of manuscripts and recensions, and each century gives us more data, not less; the Birmingham Qur'ān radiocarbon-dated to 568-645 CE, the Ṣan'ā' palimpsest that preserves the lower text, the Topkapı and Tashkent 'Uthmānic manuscripts — each is a test of whether the Quran's letters, in aggregate and in detail, are the letters of this project's corpus. They overwhelmingly are. The *ilm al-ḥarf* tradition partially survives: Ibn 'Arabī's letter-topic predictions are verified where he functions as an observer (fire→punishment, earth→creation) and falsified where he functions as a speculative metaphysician. The selective survival is the point; a tradition whose every claim passed would not be operating in reality.

Preservation-of-text itself is a future test the tradition and the discipline now conduct in concert. The Birmingham Qur'ān fragment — parchment radiocarbon-dated to 568-645 CE, held at the Cadbury Research Library — contains verses from Surahs 18, 19 and 20 in an early Ḥijāzī hand and agrees with the canonical rasm at the letter level for the verses it preserves. The Ṣan'ā' palimpsest, whose lower text was partly erased and overwritten, preserves a variant companion-codex in its underlayer; the upper layer matches the 'Uthmānic recension. The Topkapı mushaf and the Tashkent Sāmarqand mushaf, both late-first-century-AH candidates for 'Uthmānic-period copies, carry the same consonantal skeleton. What these objects together establish, at the letter level our project counts, is that the 330,709 graphemes of the canonical corpus are substantially the graphemes of the earliest recoverable witnesses; the diacritical marks, the fāṣila (verse-end) markers, and the recitation-pause signs are later scribal accretions, as expected; but the rasm underneath is stable. Each century's new manuscript discovery is in principle a test of that stability, and each century so far has not refuted it. A future opening is therefore not only forward (the hypotheses queue) but downward (new palimpsest decipherment, new carbon-dating, new character-level collation), and both directions are live.

The *ilm al-ḥarf* tradition occupies a specific place in this future. Ibn 'Arabī's letter-science cannot be rescued as a whole; the metaphysical elaborations (that each letter corresponds to a divine name, that each letter has an elemental humour, that al-Būnī's squares unlock cosmic currents) fail every rigorous test. But the phenomenological core — that letters distribute *differently* across topics in the Quran, that fire is a ق-heavy field, that earth is a ح-heavy field — partially survives: fire→punishment enrichment at $p = 5.9 \times 10^{-16}$, earth→creation at $p = 8.8 \times 10^{-12}$. The selective survival is the tradition operating as a tradition of observations, not as a system of metaphysics. Future work extends this audit to every letter $\times$ every topic under the Ibn 'Arabī rubric, and the project's preregistered framework ensures that both confirmations and falsifications will be reported. There is a

generous reading of *‘ilm al-ḥarf* — that it noticed something real about Quranic phonology before the instruments existed to measure it — and a correct reading — that it overreached where it left phenomenology for metaphysics. The generous reading is not the same as the credulous one; it is the reading that takes observations seriously while testing metaphysics critically.

One further future-opening deserves its own sentence, because the project built for it explicitly. The 22 pre-registered deep hypotheses in *findings/deep-hypotheses-queue.md* are listed in the public record precisely so that they cannot be post-hoc filtered. Every one of them commits the project in advance to a specific agent, a specific null model, a specific acceptance threshold. If the hypothesis passes, the project publishes; if it fails, the project publishes. This is the discipline the Bible-Codes controversy of the 1990s demanded and that the Code-19 tradition never adopted. Witztum, Rips and Rosenberg published the rabbi-codes paper in *Statistical Science* in 1994; McKay, Bar-Natan, Bar-Hillel and Kalai demolished it in the same journal in 1999 by demonstrating that the degree of freedom in choosing names and spellings was exactly large enough to generate the effect. The lesson the post-Bible-Codes literature absorbed is that pre-registration is not an administrative nicety; it is the only sincere answer to the multiple-comparison problem. The 22-hypothesis queue is the project's commitment to that answer. Each entry specifies the rule tuple in advance. Anyone reading the queue now can verify, five years hence, whether the project honoured its commitments.

Close with what is left standing, and what it costs to have it. The Quran is one text of 114 surahs, 6,236 verses, 77,797 real-word tokens, 330,709 letter graphemes. Its letters are drawn from a twenty-eight-letter alphabet acquired from Nabataean cursive around the fourth century CE, governing a phonology described already by Sībawayhi at the grammar's beginning and by every subsequent grammarian; its surface is now measured. Its depths were sounded by classical tradition for fourteen centuries in a vocabulary — *munāṣabat*, *mathānī*, *tarṣī*, *jinās*, *tikrār*, *radd al-‘ajuz* *‘alā al-ṣadr*, *saj*, *iltifāt*, *muḥkam-mutashābih*, *asbāb al-nuzūl*, *nasikh-mansūkh*, *muqaṭṭa‘āt*, *i‘jāz* — that the computational infrastructure inherits rather than replaces. The honest ledger, after the full audit, separates into three classes of findings. There are the survivors of honest testing: the muqaṭṭa‘āt density effect ($\chi^2 = 228.78$, $p < 10^{-15}$; Stouffer $Z = +4.48$ under a 3-gram letter-Markov surrogate); the Al-Baqarah 131-144 ring ($z = +9.69$, Bonferroni-surviving); the Q 13:28 one-verse root palindrome (jinas density 0.889, highest in the corpus); the Ash-Shams seven-verse letter-count palindrome ($p \approx 0.007$); the verse-length chronological ramp ($F = 210$); the post-Hijra Muḥammad-name monopoly; the *Rabb* chronological decline; the K-means 97%/89% Meccan/Medinan recovery; the Khawātim al-Ḥashr hapax-octet at Q 59:22-24; the Ar-Raḥmān 8+7+8+8 refrain partition encoding

the classical tafsir division; the Al-Fātiḥa nineteen-letter pivot; the Al-Kahf five-method midpoint; the Maryam triple-marked Christological pivot; the hapax-rhyme correlation at $p = 7.35 \times 10^{-29}$; the paired-opposites network with heaven/earth at $p = 1.8 \times 10^{-190}$; the eschatological speech asymmetry (saved speak *with*, damned speak *against*); the tenor-vehicle exclusivity map of parable-vehicles (13 exclusive to disbelievers, 5 exclusive to believers, 2 polyvalent: rain and garden); the Medinan > Meccan jinas density reversal (1.94×); the letter-frequency stylometric fingerprint at $|z| > 20$ on twelve letters; the 830 rhetorical questions and fewer than fifty answered; the 1,620 quotation-events and Pharaoh the only self-deifier; the 2.2% divine-name-pair verse-ending rate as a Medinan-legal marker; the *Qul* surahs being five, not four (Sūrah 72 included); the Musabbihāt set of seven cleanly recovered; the palindromic roots (*nun* palindromic AND the Nūn-muqatta'at letter AND the root of Yūnus); the graph-theoretic cohesion at $z = +36.3$. These survive.

Then there are the refutations, which are findings in their own right: *rahma* = 114 demoted to base-rate coincidence; Yūsuf sjn=12 killed by baseline; 147-triple killed by baseline; most of Khalifa Code-19 falsified; Cuypers's and Farrin's macro-rings disconfirmed at lexical level; the iron-miracle rejected as survivor bias; the ALM letter counts impossible under any single orthography; Allāh = 2,698 possible only with non-canonical verse deletion; the Zipf α non-distinctive at standard counting rule; the folk-wisdom about Meccan saj' density false. These are failures of earlier claims; they are successes of the method.

The classical tradition must stand beside them not as their scaffold but as their equal, because the classical tradition saw most of what we now count. Al-Biqā'ī's fifteenth-century *Naẓm al-Durar* described the Abraham-qibla ring our algorithm scores at $z = +9.69$. Al-Zarkashī's fourteenth-century *naw'* 52 in *al-Burhān* defined the *mutashābih al-lafẓī* phenomenon our agent quantified at 265 near-identical pairs. Al-Suyūṭī's sixteenth-century *al-Itqān* *naw'* 9 diagnosed the verse-length-by-period climb that Sadeghi re-described in 2011 and that our ANOVA replicated at $F = 210$. Al-Kirmānī's twelfth-century *Asrār al-Tikrār* catalogued 1,100+ pair instances of repetition-with-variation — the direct ancestor of our pair-extraction output. Al-Rāghib's eleventh-century *Mufradāt* was the first systematic catalog of Quranic rare words — the distant ancestor of our hapax-legomena list. Ibn al-Mu'tazz's ninth-century *Kitāb al-Badī'* founded *radd al-'ajuz 'alā al-ṣadr* — the chiasmus category the Q 13:28 palindrome inhabits. Al-Farrā'ī's ninth-century *Ma'ānī al-Qur'ān* located each grammatical phenomenon in the text. Sibawayhi's late-eighth-century *al-Kitāb* defined the phonetic classes our phonaesthetics agent used. Al-Khalīl's late-eighth-century *Kitāb al-'Ayn* gave the lexicographic frame. Each of these is an ancestor our work cites by name. *The genealogy is explicit; the novelty is application, not conception.*

And there are the genuinely novel observations this project contributed without classical precedent — the eleven whose classical category is empty: the McKay denominator of 2,817 as the mandatory baseline for any future balance claim; the graph-theoretic bridge roots *k-h-l-q* (create) and *q-l-b* (heart); the letter-frequency stylometric fingerprint; the chronological verse-length metric at ANOVA-level rigor; the Bonferroni-corrected ring catalog with a 57,996-window family size; the compression-detectable refrain of Ar-Raḥmān; the hapax-rhyme correlation at 7.35×10^{-29} ; the five-mode rhyme-break taxonomy; the paired-parable antithesis pattern; the 4-tier dialogical speech-event architecture; the thirteen-layer structural vocabulary.

A fourth class of observation — the observation the project makes about itself — needs to be named here, because the register of the work depends on it. The project's invariants: the Quran is one text (no "editions" framing); every numerical claim carries a rules tuple (per methodology.md §9); every novel finding passes at least two null models before being called significant; prior art is WebSearched and documented regardless of novelty; failures are reported with equal prominence to successes; intelligence over mechanics — classical tafsīr and balāgha are cross-referenced on every substantive finding. The invariants are the project's spine and they are the reason some findings that looked promising in month one did not survive to month three. The *rahma*=114 demotion, the Yūsuf-sjn=12 killing, the 147-triple killing, the ALM letter-count failures, the Farrin macro-ring disconfirmation — each is a finding the project would have published in any less disciplined register, and each was withdrawn because the invariants demanded it. The withdrawn findings are listed with the same prominence as the surviving ones. That is not apology; that is method. The apologetic tradition has been characterised by a one-way publication selection that our predecessors did not pre-register against and that therefore generated a literature of survivors only. We publish the denominator. *The denominator is the method.*

One further meta-observation completes the picture: where findings converge. Convergence is the structural vindication of a claim by independent methods operating on independent signals. Al-Kahf at the midpoint of the book is converged by five such methods — word-midpoint, letter-midpoint, longest alif monorhyme, two Bonferroni-surviving sub-rings, densest cross-surah rhyme link with Al-Jinn. Al-Baqarah 131-144 is converged by six methods — chiasitic audit, middle-ayah (wasat), jinās, graph-theory hub, saj-rhyme qibla-shift, surah-boundaries inclusio. Khawātīm al-Ḥashr is converged by seven — divine-name density, hapax-name concentration, 49-word/216-letter numerology, twin-opener device, the *al-husnā* meta-statement, recapitulation of Al-Fātiḥa's opening name-sequence, and the apophatic-kataphatic diptych with Āyat al-Kursī. When five or six or seven independent methods identify the same pericope as a structural hotspot,

the probability that every one of them happens to be wrong in the same place approaches zero. That is the shape of converged finding, and the converged findings are where the project is most confident.

What this project does not do is pretend that the numbers, by themselves, secure the theological claim. The Quran's religious status is not a matter this essay arbitrates. What the essay can say is narrower: the Quran is an exceptionally rhetorically dense, theologically coherent, linguistically rich text whose specific numerological aura is mostly manufactured by retrospective selection, and whose actual structural signatures are more subtle, more local, and more interesting than the apologetic tradition ever claimed. The text does not need the Code-19 decoration. What it has — the rings that survive Bonferroni, the hapaxes lodged at rhyme, the refrain that encodes its own section-division, the midpoint verse that names the jinn, the palindrome that enacts rest, the self-naming *mathānī* that tells you the mode it is in — is more than any numerology could have designed. And the classical tradition, in the work of al-Zarkashī and al-Suyūṭī and al-Biqā'ī and al-Rāghib and al-Kirmānī and Ibn al-Mu'tazz and Ibn 'Arabī, had already named almost all of it. The project's contribution, when the fourteen centuries and the algorithms are put together, is a single braided thing: the classical vocabulary made testable, and the negative results published with the same prominence as the positive.

The single most compact way to hold all of this together is the sentence Q 21:107 offers about itself: *wa-mā arsalnāka illā raḥmatan li-l-‘ālamīn* — "and we have not sent you except as a mercy to the worlds". The text that uses the word *raḥma* 114 times, with 114 surahs holding those uses, describes its own purpose with the same word. Whether the numerological coincidence is base-rate coincidence (as the cross-baseline test demonstrated) or significant (as the family-level robustness preserves narrowly), the *thematic* coincidence is not contingent on the count. The Quran says what it intends to be: a mercy. It repeats the word for that intention more than a hundred times in its body. And the computational infrastructure we have added, far from being an external instrument forced on the text, turns out — in every surviving finding — to be something the text's own self-descriptive vocabulary had already anticipated. *Mathānī* was the word; the ring was the structure; the refrain was the rhythm; the palindrome was the *radd al-‘ajuz*; the hapax was the *gharīb*; the pivot was the *wasat*; the rhyme-break was the *khurūj min al-fāṣila*. Every tool this project built had a classical name, and the classical name was given to it by scholars who could not run a null model but could describe the phenomenon with the precision of natural observers.

Close on the rule the project repeats in its own documents and will repeat to every successor: *a number is not a claim until its rules tuple is disclosed*. That sentence distinguishes this work from every apologetic miracle-listicle that has ever been published

about the Quran; it is also the discipline every future attempt — apologetic, skeptical, or merely curious — must inherit. The Quran's ambition we cannot exhaust. The surface we have measured. The depths the tradition has been sounding for fourteen centuries. The openings still available are as many as the hypotheses we have not yet run. And every number that will ever be attached to the Quran, in praise or in critique, is answerable to the same disclosure: under which orthography, which segmentation, which basmala policy, which verse-numbering, which null model? *That* is the sentence the project asks every reader to carry out of it. The Quran is one text — 114 surahs, 6,236 verses, 77,797 words, 330,709 letters — and a number is not a claim until its rules tuple is disclosed.